Clustering 1

Introduction to Data Mining

Seoul National University

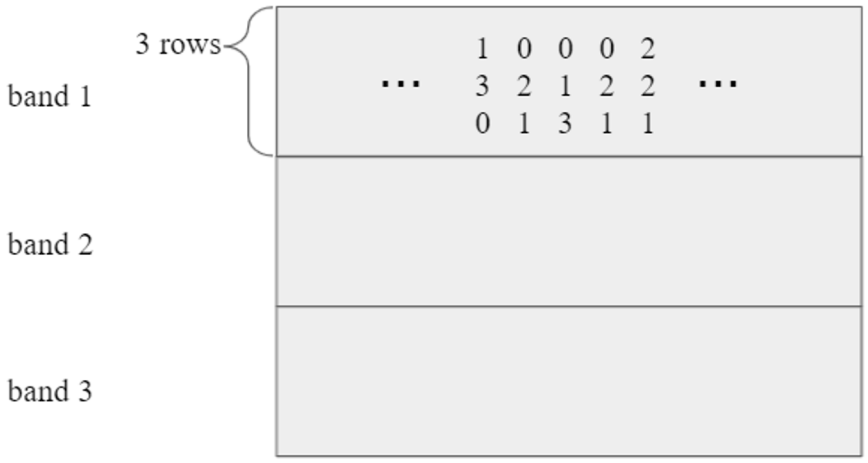
Spring 2026

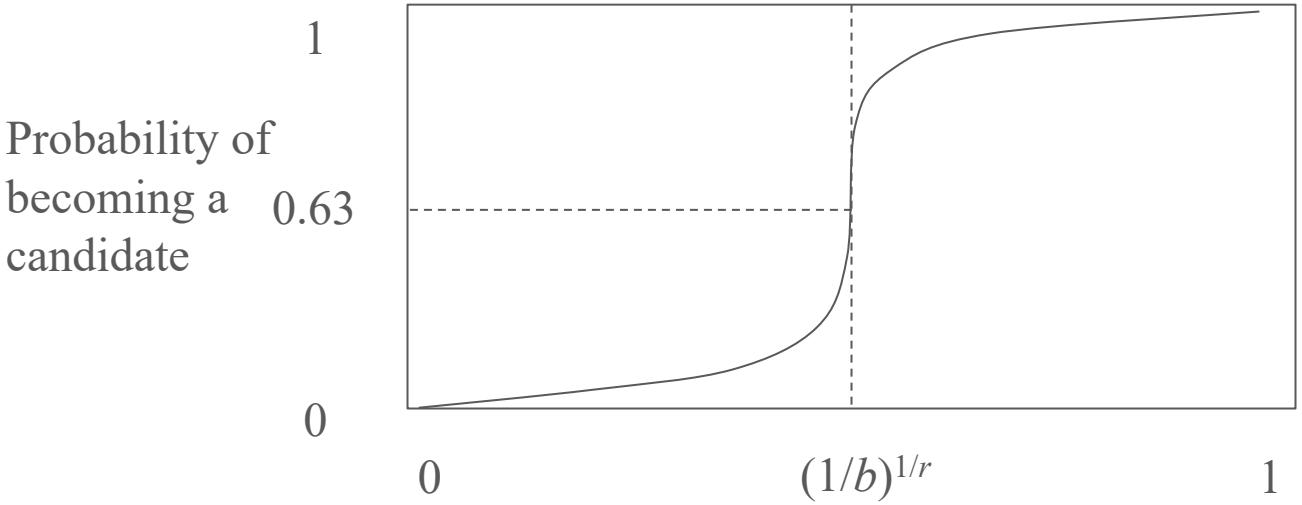
Announcements

- **Assignments:**
  - HW1 is posted (deadline: April 9).
- **Languages:**
  - Assignments allow only English.
  - Exams allow both English and Korean.
- **Using eTL:**
  - You can ask any questions regarding lectures on the Lectures board.

Recap: LSH

- Locality Sensitive Hashing (LSH)
  - S-curves
- Theory of LSH:
  - Locality-sensitive families
  - AND and OR constructions
- Other distance measures:
  - Random hyperplanes





Outline

1. **<u>Clustering</u>**
2. Hierarchical Clustering
3. K-means Clustering

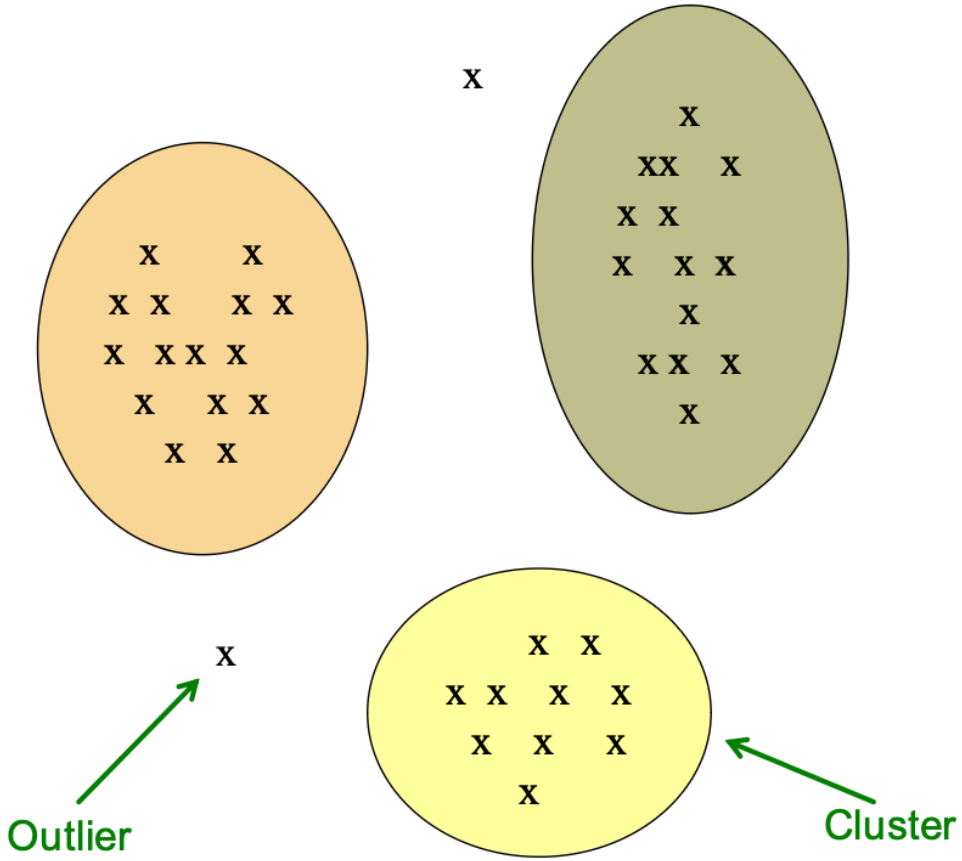
Clustering

- The process of grouping a collection of points into **clusters**.
  - Members of the same cluster are close/similar to each other.
  - Members of different clusters are dissimilar.
- We focus on cases when
  - Data is very large.
  - Space is high-dimensional.
  - Space is non-Euclidean.
  - Similarity can be defined using any distance measure.

Distance Measures

- **Distance measure** gives a distance between two points.
  - Recall the 4 requirements for a distance measure.
- Popular distance measures in a Euclidean space include:
  - Euclidean (L2) distance: $\sqrt{\sum_i (x_i - y_i)^2}$.
  - Manhattan (L1) distance: $\sum_i |x_i - y_i|$.
  - L0 distance: $\sum_i \left[1 \text{ if } x_i \neq x_y \text{ else } 0\right]$.
  - L-Infinity distance: $\max_i |x_i - y_i|$.

Example: Clusters and Outliers



Source: Stanford CS246 (2022)

Example: Documents

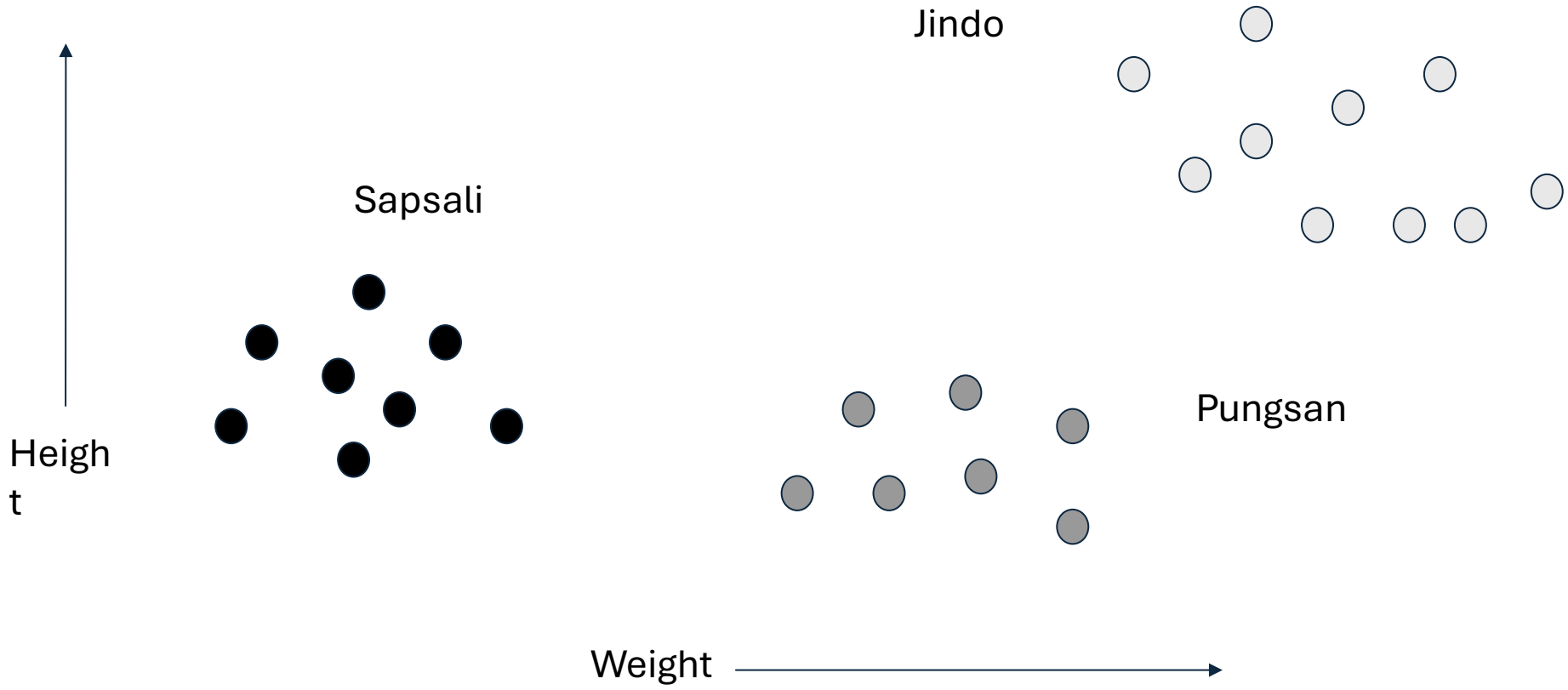
- Represent a document by a vector $(x_1, x_2, \cdots, x_k)$.
  - $x_i = 1$ iff the i-th word appears in the document.
- **Idea:** Documents with similar words are about the same topic.
- **Task:** Find clusters of documents.
  - Different from finding similar pairs, which returns document *pairs*.

Cosine, Jaccard, and Euclidean

- We have a choice when we think of documents as sets of words:
  - **Sets as vectors:** Measure similarity by the cosine distance.
  - **Sets as sets:** Measure similarity by the Jaccard distance.
  - **Sets as points:** Measure similarity by the Euclidean distance.

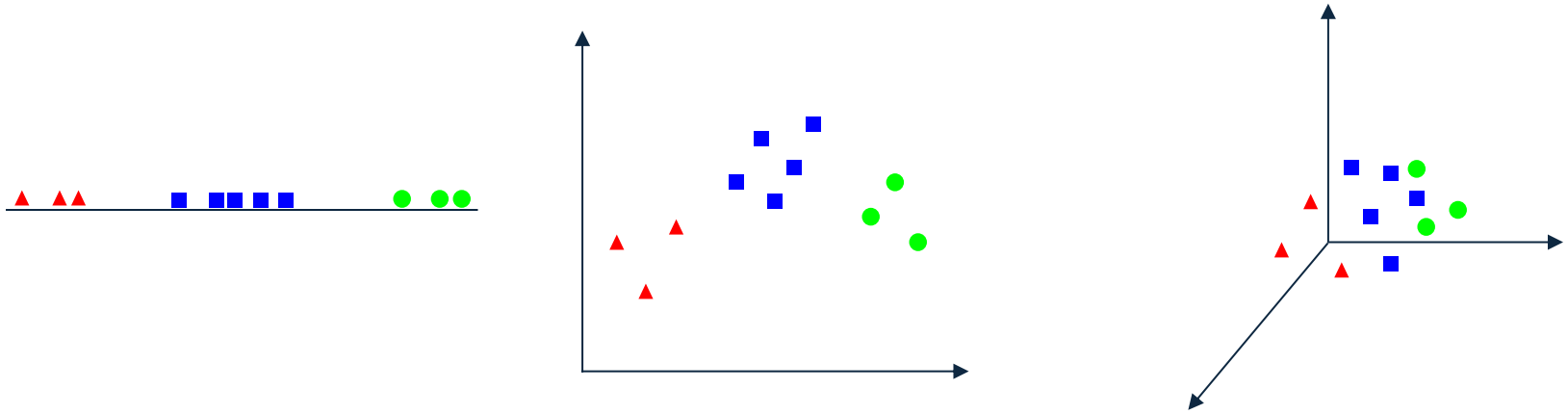
Clustering in Low-Dimensional Space

- Clustering in two dimensions looks easy.



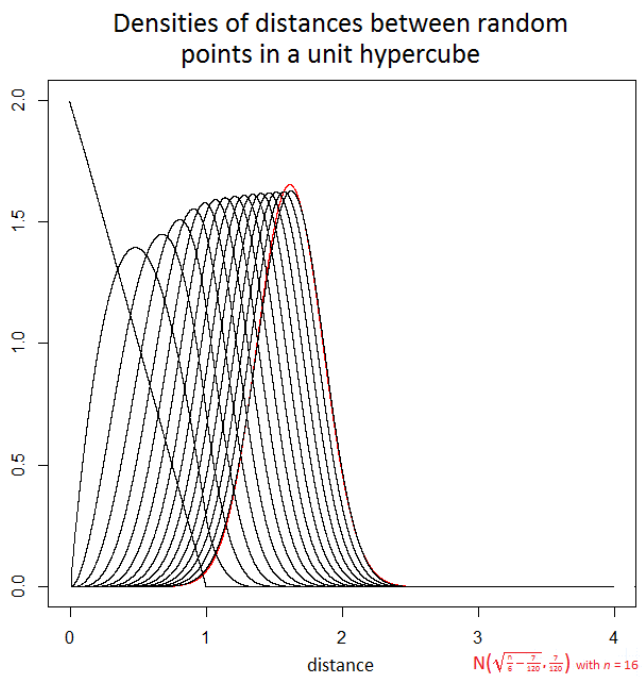
The Curse of Dimensionality

- High-dimensional spaces have the **curse of dimensionality**.
  - Almost all pairs of points are equally far away from one another.



The Curse of Dimensionality

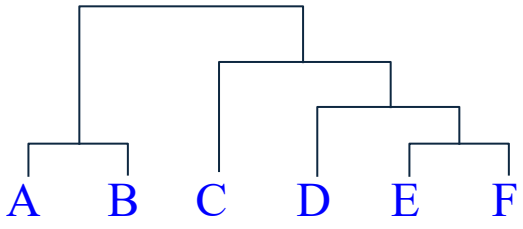
- Consider two random points in the n-dimensional unit hypercube.
- The variance of their distance goes to a constant when $n \to \infty$.
  - But the possible range is $[0, \sqrt{n})$.
- Suppose that $n = 2500$.
  - The distance can be anything in $[0, 50)$.
  - However, most distances are in $(20, 21)$.

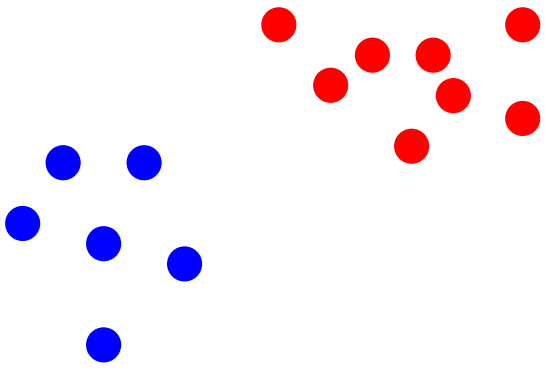


Source: https://math.stackexchange.com/questions/1976842/how-is-the-distance-of-two-random-points-in-a-unit-hypercube-distributed

Clustering Strategies

- **Hierarchical:**
  - **Agglomerative** (bottom up)
    - Initially, each point is a cluster.
    - Repeatedly combine the “nearest” clusters into one.
  - **Divisive** (top down)
    - Start with one cluster and recursively split it.
- **Point assignment:**
  - Maintain a set of clusters.
  - Points belong to the “nearest” clusters.





Hierarchical vs. Point-Assignment

- (Left) Point assignment is good when clusters have nice shapes.
- (Right) Hierarchical methods can win when the shapes are weird.
  - Note that both clusters have essentially the same centroid.

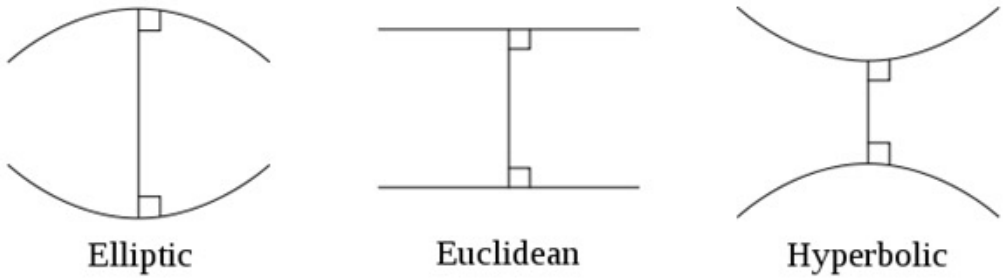
Source: Stanford CS246 (2022)

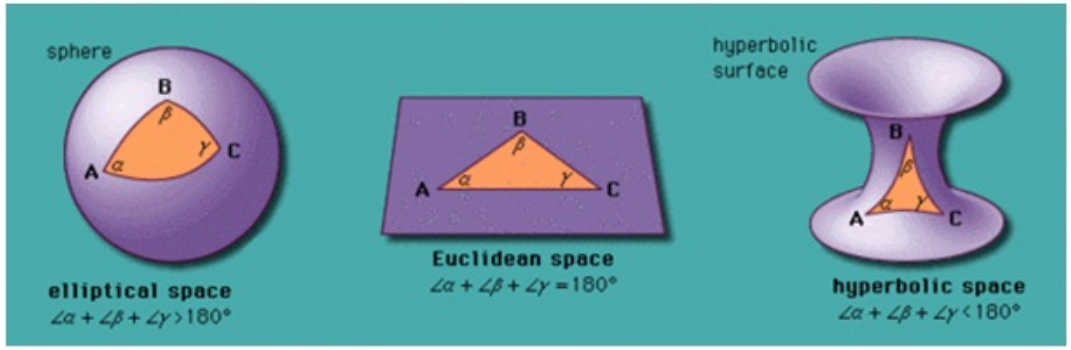
Clustering Strategies

- **In Euclidean:**
  - Points are vectors of real numbers, i.e. coordinates.
  - It is possible to summarize a set of points as a **centroid**.
  - Distance measure: L2 norm, L1 norm, etc.
- **In non-Euclidean:**
  - There is no notion of “location” or centroids.
  - We should summarize a collection of points differently.
  - Distance measures: Jaccard, Hamming, cosine, etc.

Non-Euclidean Space

- Finding the center of points is difficult in a non-Euclidean space.
  - E.g., a simple average of three points will not lie on the sphere.





Source: LinkedIn

Clustering Strategies

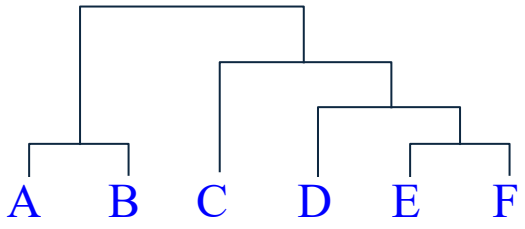
- **In-memory clustering:** Easier if the data fit in memory.
  - Example: K-means
- **Large-data clustering:** Load one batch of data at a time.
  - Cluster them in memory and keep summaries of clusters.
  - Example: BFR and CURE

Outline

1. Clustering
2. **<u>Hierarchical Clustering</u>**
3. K-means Clustering

Hierarchical Clustering

- **Key operation:** Repeatedly merge two “nearest” clusters.
- **Three important questions:**
  1. How to represent a cluster?
  2. How to determine the *nearness* of clusters?
  3. When to stop merging clusters?



Hierarchical Clustering: Euclidean

- In the Euclidean case, our answers are straightforward.
- **Q1:** How to represent a cluster of many points?
  - As we merge clusters, we represent the “location” of each cluster by
    - Its centroid = average of its (data) points.
- **Q2:** How to determine the nearness of clusters?
  - Measure cluster distances by the distances between their centroids.
  - Merge two clusters with the shortest distance.

Example: Hierarchical Clustering

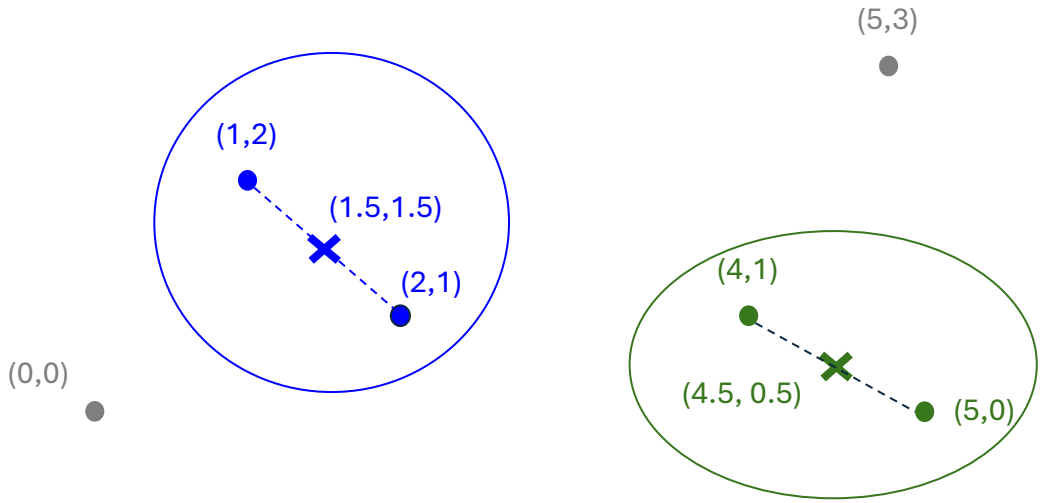
Data:

● = data point

✕ = centroid

Dendogram

Example: Hierarchical Clustering



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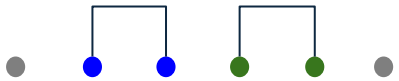
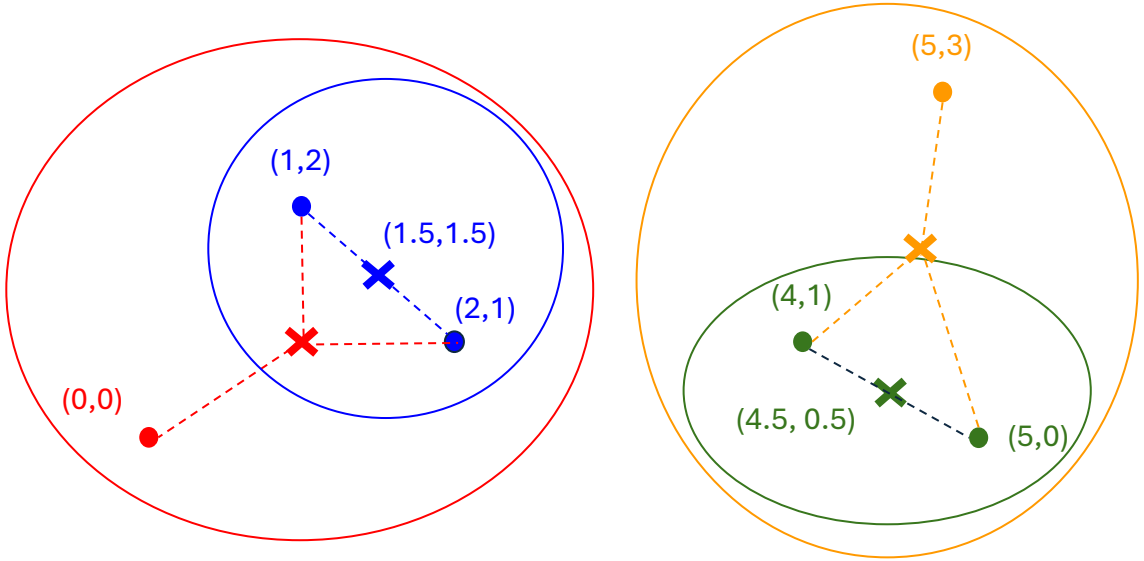


Dendogram

Example: Hierarchical Clustering



Data:

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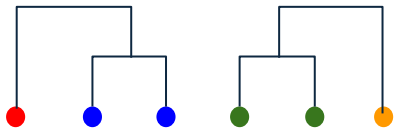
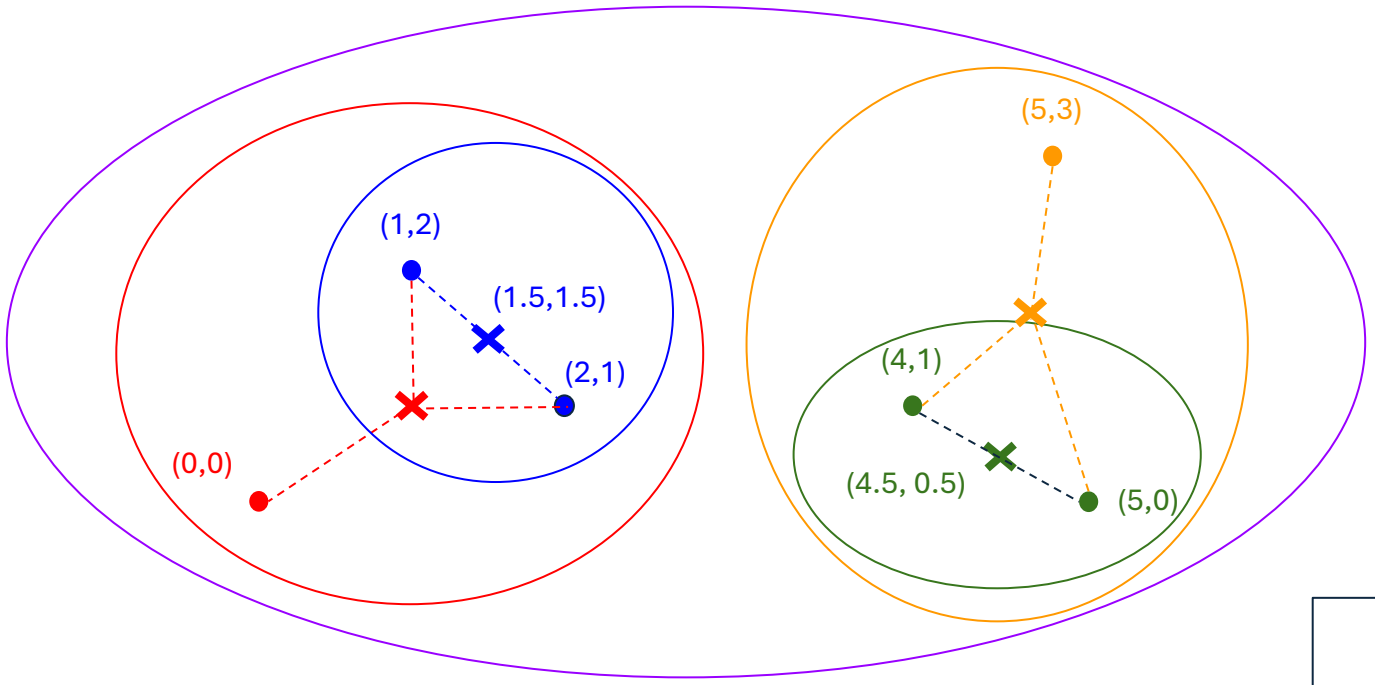


Dendogram

Example: Hierarchical Clustering



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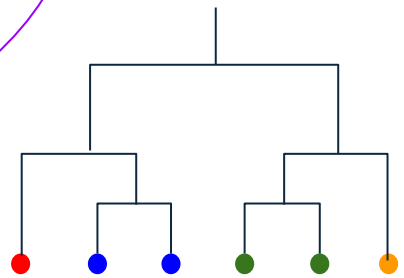
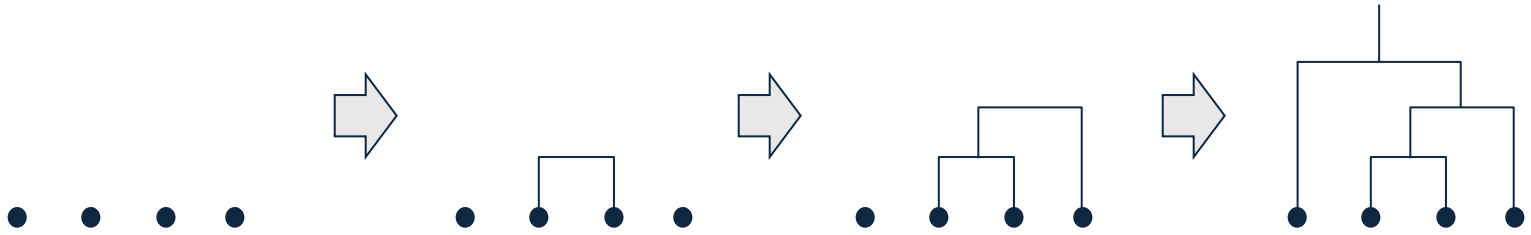


Dendogram

Efficiency of Hierarchical Clustering

- **Basic algorithm:**
  - At each step, compute distances between pairs of clusters.
  - Running time is $O(n^3)$, since $n^2 + (n-1)^2 + \ (n-2)^2 + \cdots$



Efficiency of Hierarchical Clustering

- More efficient implementation uses a **priority queue**.
  - Store a collection of prioritized elements.
  - The overall algorithm is $O(n^2 \log n)$, which is better than $O(n^3)$.
- Why is it better?
  - We avoid re-computing the distances computed earlier.
  - See textbook for details.

Hierarchical Clustering: Non-Euclidean

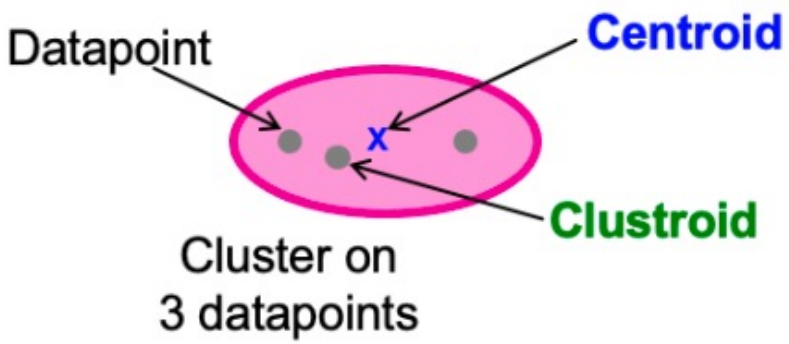
- In a non-Euclidean case:
  - The only “locations” we can talk about are the points themselves.
- **Three possible approaches** regarding the questions:
  - **Q1:** How to represent a cluster of many points?
  - **Q2:** How to determine the nearness of clusters?
  - **Note:** These approaches can also be used in the Euclidean case.

Non-Euclidean Approach 1

- **Q1:** How to represent a cluster of many points?
  - **Pick a clustroid** = (data) point "closest" to all other points.
  - Possible meanings of "closest":
    - Smallest maximum / average distance to all other points.
- **Q2:** How to determine the nearness of clusters?
  - Treat clustroid as if it were centroid.

Centroids vs. Clustroids

- **Centroid** is the average of all (data) points in the cluster.
  - This means the centroid is an “artificial” point.
- **Clustroid** is an **existing point** that is “closest” to all other points.



Source: Stanford CS246 (2022)

Non-Euclidean Approach 2

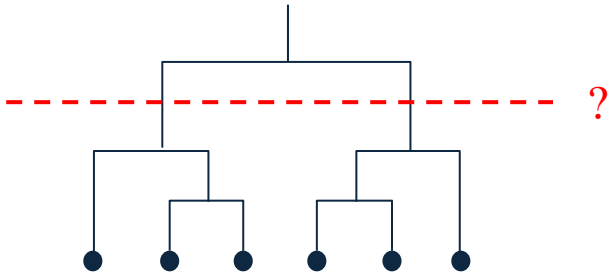
- **Q1:** How to represent a cluster of many points?
  - As the collection of points in the cluster.
- **Q2:** How to determine the nearness of clusters?
  - **Define inter-cluster distance:**
    - Minimum of the distances between any two points, one from each cluster.
    - Average distance of all pairs of points, one from each cluster.

Non-Euclidean Approach 3

- **Q1:** How to represent a cluster of many points?
  - As the collection of points in the cluster.
- **Q2:** How to determine the nearness of clusters?
  - **Merge clusters whose union is most cohesive.**
  - Possible notions of cohesion (the smaller, the more cohesive):
    - Diameter of the merged cluster = Maximum distance between points.
    - Average distance between all points.
    - Inverse density of the cluster = Diameter or average distance / # of points.

When to Stop?

- When do we stop merging clusters?
  - If the diameter of the merged cluster exceeds threshold.
  - If the density of the merged cluster is below threshold.
  - If there is evidence that the merged cluster yields a bad cluster.
    - E.g., the average diameter suddenly increases.



Pop Quiz

- Perform a hierarchical clustering on the points below.
  - The distance between two clusters is:
    the maximum of the distances between any two points.
- 1, 4, 9, 16, 25, 36

Outline

1. Clustering
2. Hierarchical Clustering
3. **<u>K-means Clustering</u>**

k-means Algorithm

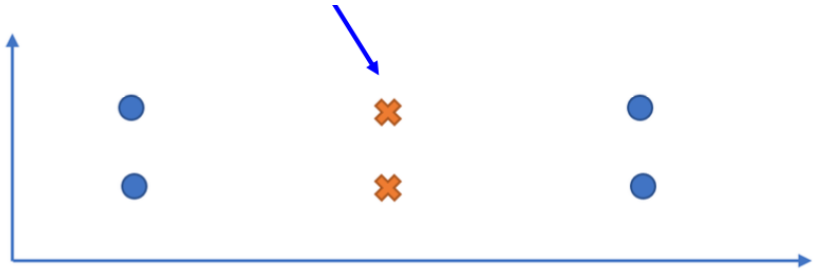
- Best known family of clustering algorithms for point assignment.
- **Given** Euclidean space/distance and k = number of clusters.
  - Possible to deduce k by trial and error.
- **Find** cluster centers that minimize the sum of squared distances.
  - From each point to its cluster center.

k-means Algorithm

- **Initialize** clusters by picking k centers.
- **Until convergence:**
  1. For each point, assign it to the cluster whose centroid is the closest.
  2. After all points are assigned, update the centroids of the k clusters.
     - As the average of data points within each cluster.
- **Convergence** means that points don't move between clusters.

Shortcoming of k-means

- Convergence of k-means heavily depends on the initial centroids.
- It can perform arbitrarily badly in such a case:

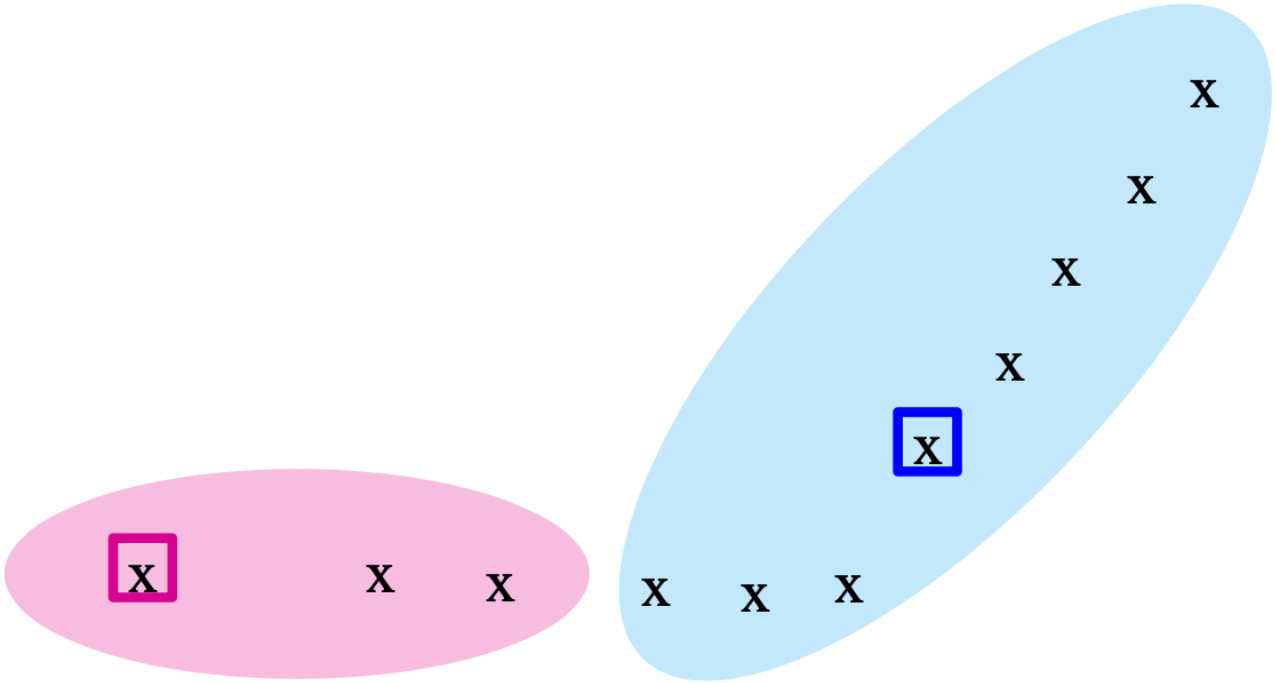


Source: Stanford CS246 (2022)

k-means++

- **Goal:** Pick points that are likely to lie in different clusters.
- **Basic idea:**
  - Pick a small sample of points S, cluster them, and use their centroids.
    - Any algorithm can be used for the clustering.
  - Sample size $|S|$ is proportional to $k \times \log n$, where n is the data size.
- **How to pick sample points:**
  - Visit points in a random order.
  - Add each point p to the sample with a probability proportional to $D(p)^2$.
    - $D(p)$ = Distance between p and the nearest already picked point.

Example: Assigning Clusters



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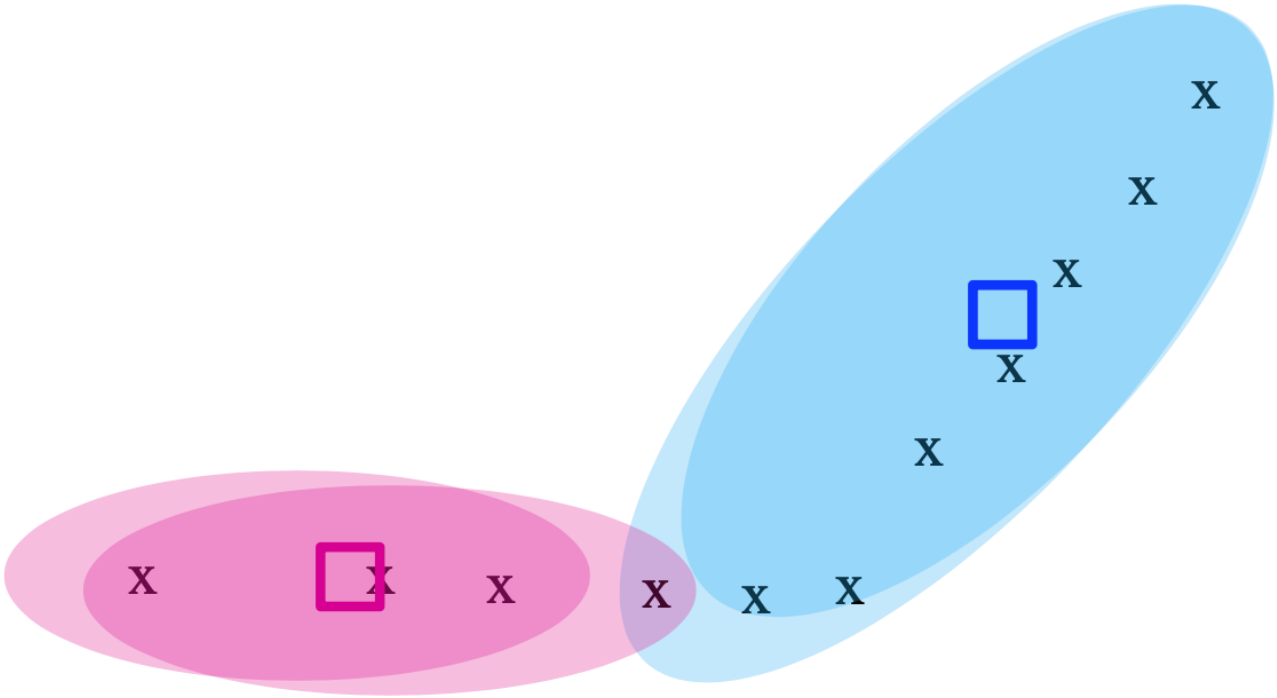

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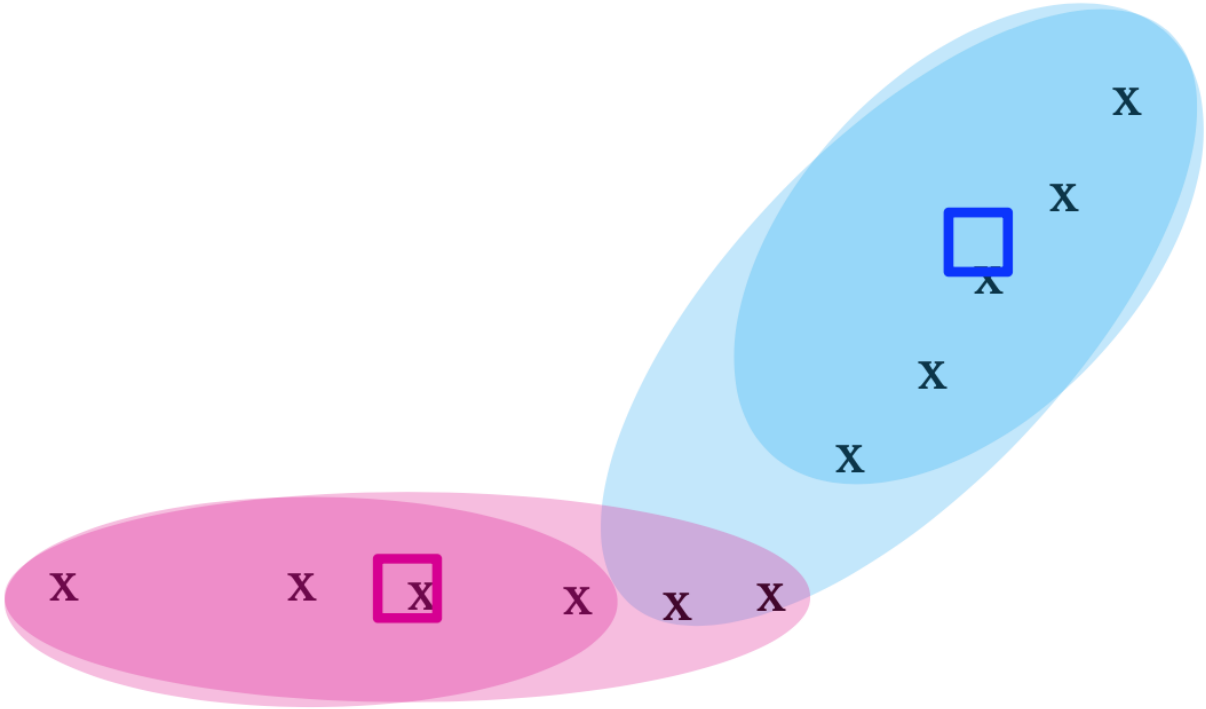
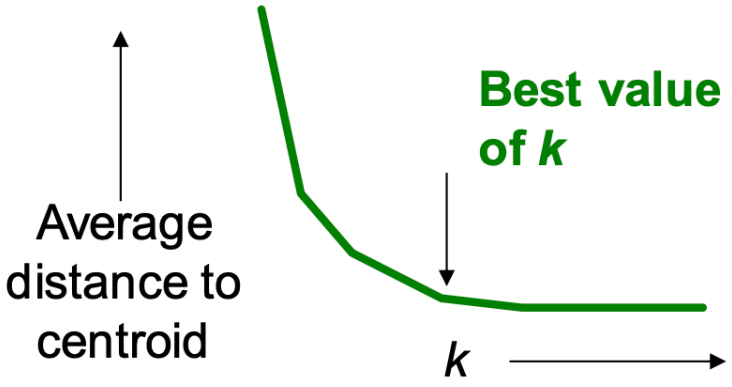
Example: Assigning Clusters



Source: Stanford CS246 (2022)

Getting the k Right

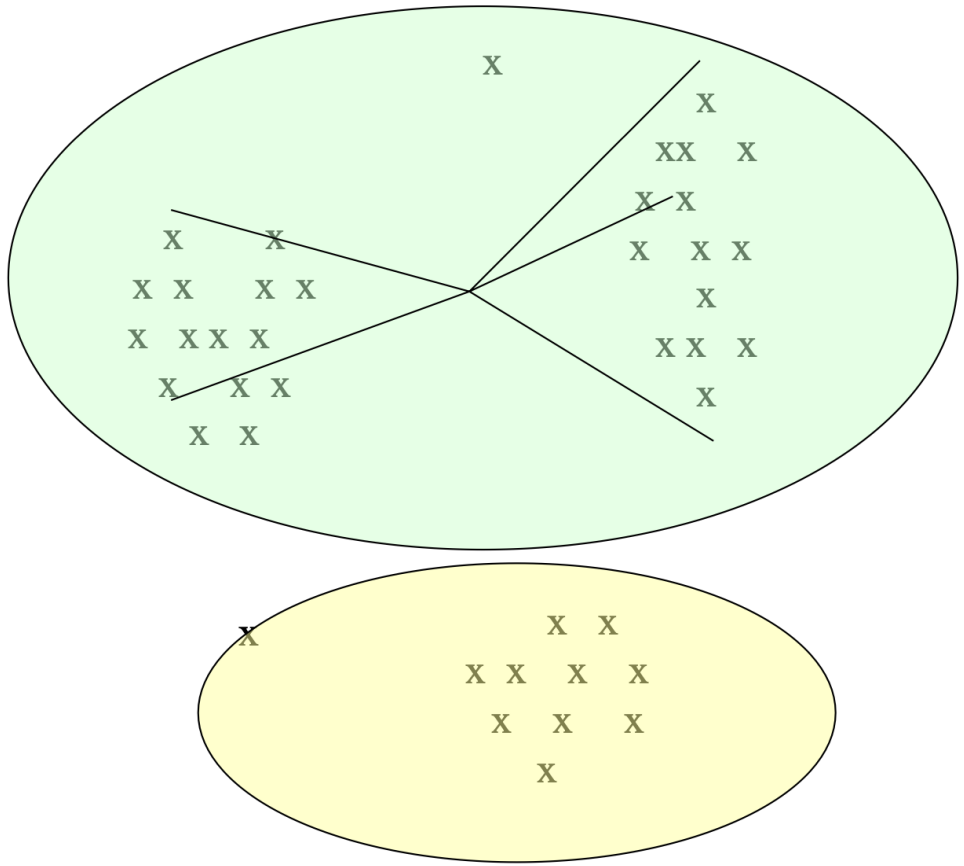
- **How to select k?**
  - Try different k.
  - Look at the change in the average distance to centroid as k increases.
  - Average falls rapidly until right k, then changes little.



Source: Stanford CS246 (2022)

Example: Picking k

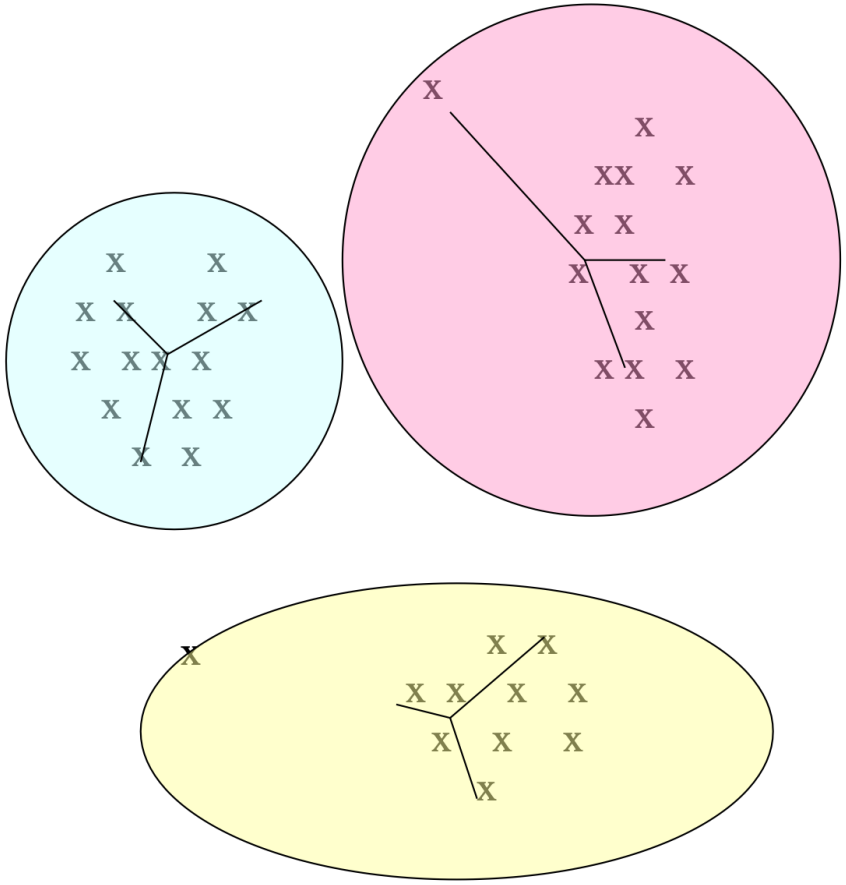
Too few;
many long
distances
to centroid



Source: Stanford CS246 (2022)

Example: Picking k

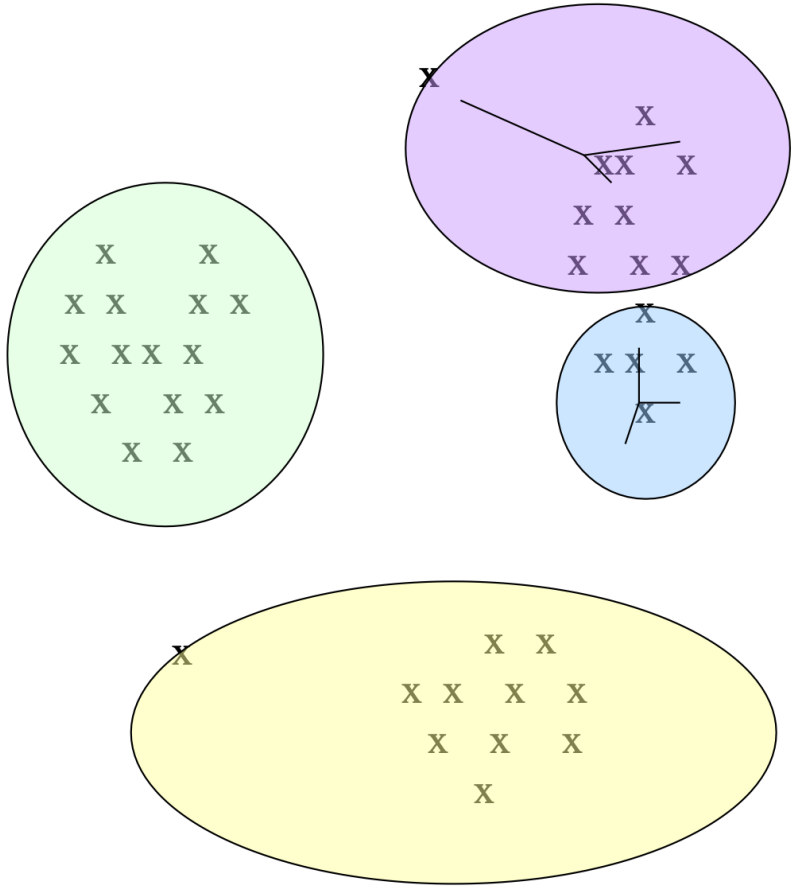
Just right;
distances
rather short



Source: Stanford CS246 (2022)

Example: Picking k

Too many;
little improvement
in average
distance



Source: Stanford CS246 (2022)

Summary

1. Clustering
   - Curse of dimensionality
   - Clustering strategies
2. Hierarchical Clustering
   - Euclidean vs. non-Euclidean
   - Centroids vs. clustroids
3. K-means Clustering
   - k-means++
   - Selection of k